

Supplemental Appendix: Cleansing Oil

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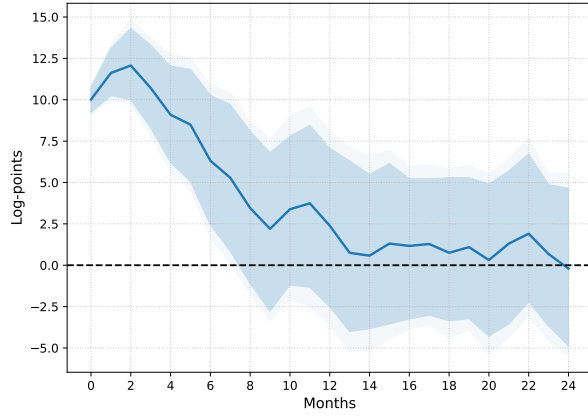
A Empirical Validation of Oil Shock Identification

Figure A.1 compares the impulse response functions generated from estimating equation (1) for the real WTI oil price and OECD+6 industrial production index using both the oil news shock of Känzig (2021) and the structural oil supply shock of Baumeister and Hamilton (2019), both normalised to a 10% increase in oil prices on impact. The two series generate virtually identical price dynamics: oil prices rise sharply on impact and decay to zero within 8 months, so agents face the same price signal regardless of which shock is used. The two series diverge, however, in their implied economic activity effects. The news shock delivers the expected contractionary response: industrial production falls on impact and remains depressed for the full 2-year horizon. The structural supply shock, by contrast, yields a positive and statistically significant increase in industrial production over the 2001–2024 sample — the opposite of what theory and the broader literature predict.

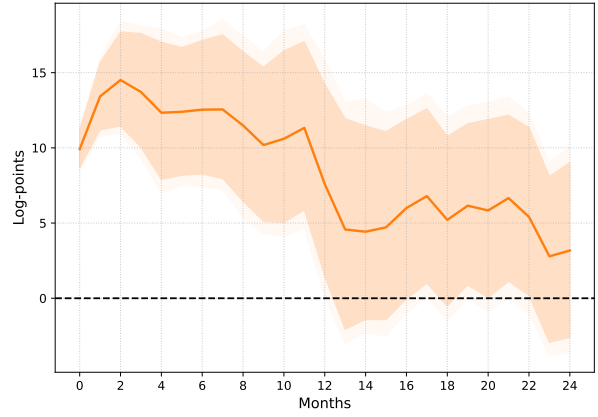
There are many reasons why this could be happening. One is the contamination of the shocks, as argued by Mori and Peersman (2024). Nevertheless, including more financial controls does not solve the problem. When using a shorter sample starting in 2010, consistent with the panel data, the problem becomes less severe, though the response is more delayed than what has been documented in the literature. The results are plotted in figure A.2. The oil news shock, by contrast, generates coherent dynamics across the samples used in the main analysis: starting in 2001, when

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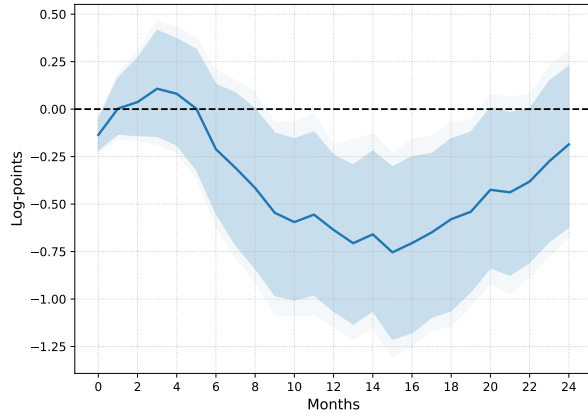
Figure A.1: Oil News Shock vs. Structural Supply Shock: Price and Activity Responses



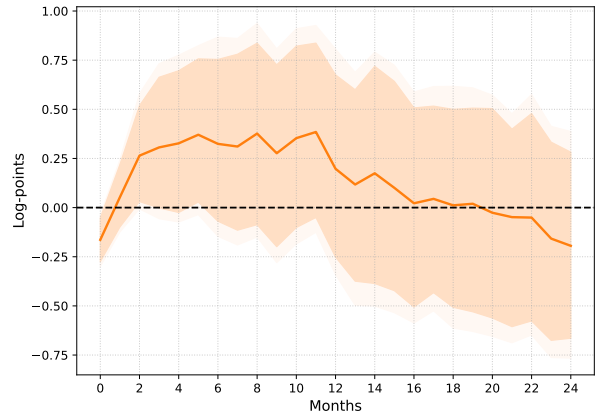
(a) Real WTI — Oil News Shock



(b) Real WTI — Structural Supply Shock



(c) Industrial Production — Oil News Shock

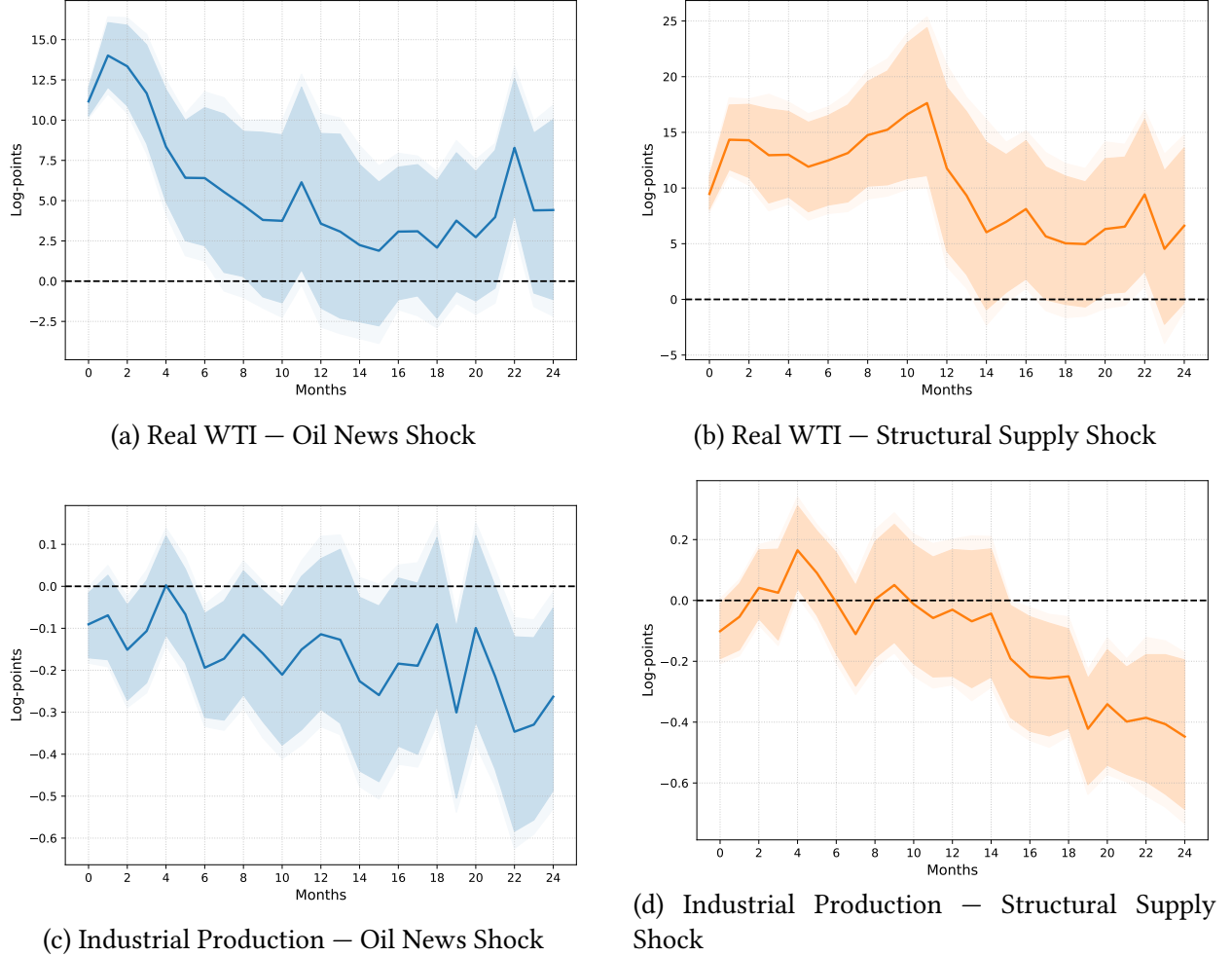


(d) Industrial Production — Structural Supply Shock

Notes: Impulse responses to each shock series, normalised to a 10% increase in oil prices on impact. Left column: oil supply news shock of Känzig (2021). Right column: structural oil supply shock of Baumeister and Hamilton (2019). Top row: real WTI oil price (log-points). Bottom row: OECD+6 industrial production (log-points). Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

the emissions sample begins, or in 2010, when panel coverage broadens significantly, the shock induces the same response in industrial production and oil prices, albeit with different persistence. I therefore use the oil news shock throughout to ensure consistency.

Figure A.2: Oil News Shock vs. Structural Supply Shock: Price and Activity Responses — 2010 Sample

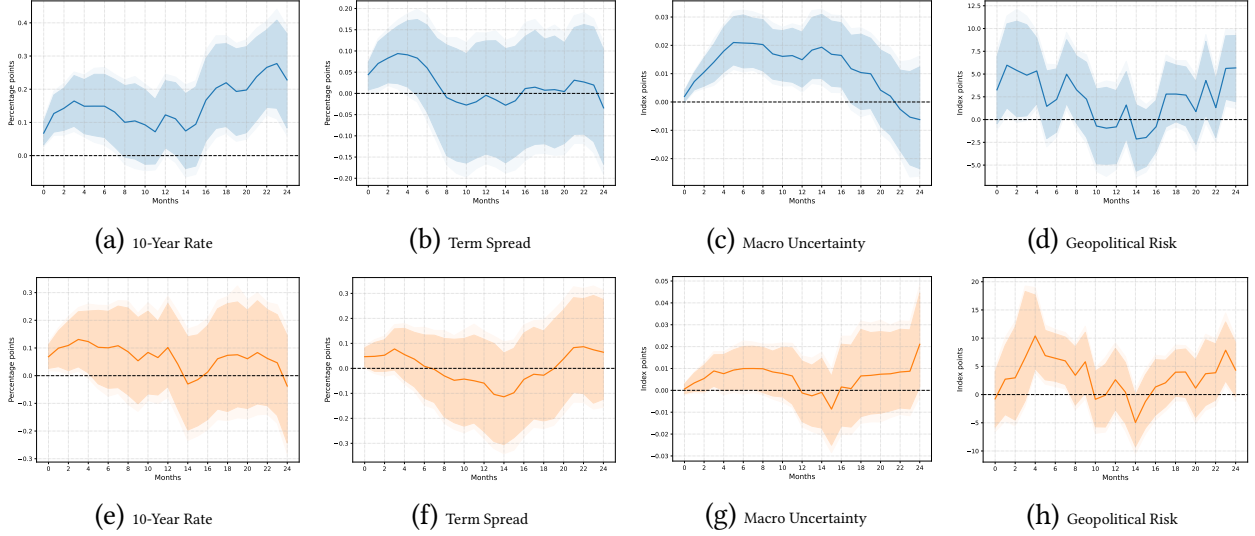


Notes: Impulse responses to each shock series, normalised to a 10% increase in oil prices on impact. Left column: oil supply news shock of Känzig (2021). Right column: structural oil supply shock of Baumeister and Hamilton (2019). Top row: real WTI oil price (log-points). Bottom row: OECD+6 industrial production (log-points). Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded. Shocks: January 2010 – December 2024; outcome data extends to December 2025.

I also compare the dynamics of the other macroeconomic indicators used in subsection 4.2 — the 10-year nominal interest rate, the term spread, macroeconomic uncertainty, and the geopolitical risk index — induced by the oil news shock and the structural oil supply shock. Figure A.3 shows that, unlike industrial production, the induced impulse response functions are similar

across both shocks.

Figure A.3: Oil News Shock vs. Structural Supply Shock: Financial and Uncertainty Responses



Notes: Impulse responses to each shock series, normalised to a 10% increase in oil prices on impact. Top row: oil supply news shock of Känzig (2021). Bottom row: structural oil supply shock of Baumeister and Hamilton (2019). Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

B Construction of the CO_2 -Equivalent Concentration Index

The time-series analysis uses a CO_2 -equivalent (CO_2e) concentration index that aggregates the atmospheric burden of 4 major greenhouse gases— CO_2 , CH_4 , N_2O , and SF_6 —into a single, consistent metric. Monthly global-mean concentrations of CO_2 (ppm), CH_4 (ppb), N_2O (ppb), and SF_6 (ppt) are obtained from NOAA.

Radiative forcing. Each gas contributes to total radiative forcing (RF) relative to its pre-industrial (1750) reference concentration. I use the simplified stratospheric-temperature-adjusted RF (SARF) expressions of Etminan et al. (2016) as adopted in Intergovernmental Panel on Climate Change (IPCC) (2023). Let C , M , N , S denote current concentrations of CO_2 (ppm), CH_4 (ppb), N_2O (ppb), and SF_6 (ppt), with pre-industrial reference values $C_0 = 278.3$, $M_0 = 729.2$, $N_0 = 270.1$,

$S_0 = 0$. The individual SARF terms (W m^{-2}) are:

$$\Delta F_{\text{CO}_2}(C, N) = \left[\alpha(C) + c_1 \sqrt{N} \right] \ln\left(\frac{C}{C_0}\right) \cdot e_1, \quad \alpha(C) = d_1 + a_1(C - C_0)^2 + b_1(C - C_0) \quad (1)$$

$$\Delta F_{\text{CH}_4}(M, N) = \left[a_2 \sqrt{M} + b_2 \sqrt{N} + d_2 \right] \left(\sqrt{M} - \sqrt{M_0} \right) \cdot e_2 \quad (2)$$

$$\Delta F_{\text{N}_2\text{O}}(C, M, N) = \left[a_3 \sqrt{C} + b_3 \sqrt{N} + c_3 \sqrt{M} + d_3 \right] \left(\sqrt{N} - \sqrt{N_0} \right) \cdot e_3 \quad (3)$$

$$\Delta F_{\text{SF}_6}(S) = \omega (S - S_0) \quad (4)$$

The scalars e_1, e_2, e_3 are the stratospheric temperature adjustment factors that convert instantaneous RF to SARF. Total forcing is $\Delta F_{\text{total}} = \sum_g \Delta F_g$. Parameter values are listed in table B.1.

Table B.1: SARF parameter values (Etminan et al. 2016; Intergovernmental Panel on Climate Change (IPCC) 2023). [†]Units of $\text{W m}^{-2} \text{ppt}^{-1}$.

Gas	Parameter	Value
CO_2	a_1	-2.48×10^{-7}
	b_1	7.59×10^{-4}
	c_1	-2.15×10^{-3}
	d_1	5.2488
	e_1 (adj. factor)	1.05
CH_4	a_2	-8.96×10^{-5}
	b_2	-1.25×10^{-4}
	d_2	0.045194
	e_2 (adj. factor)	0.86
N_2O	a_3	-3.42×10^{-4}
	b_3	2.546×10^{-4}
	c_3	-2.44×10^{-4}
	d_3	0.12173
	e_3 (adj. factor)	1.07
SF_6	$\omega^\dagger$	5.67×10^{-4}

CO_2 -equivalent concentration. Total forcing ΔF_{total} is inverted to a CO_2e concentration C_{eq} defined as the level of CO_2 that, holding N_2O at its pre-industrial value N_0 , would produce the

same forcing as all 4 gases combined:

$$\Delta F_{\text{CO}_2}(C_{\text{eq}}, N_0) = \Delta F_{\text{total}}. \quad (5)$$

Since equation (1) is not analytically invertible in C , C_{eq} is solved for numerically each period using Brent's method over the interval $[C_0, 5000]$ ppm.

C Wild Bootstrap Procedure

Confidence bands for the time-series local projections are computed using a residual-based wild bootstrap (Liu 1988). For each horizon h , let $\hat{\beta}_h$ denote the OLS estimate of equation (1) and let $\hat{y}_t$ and $\hat{\varepsilon}_t$ denote the corresponding fitted values and residuals. The procedure is as follows.

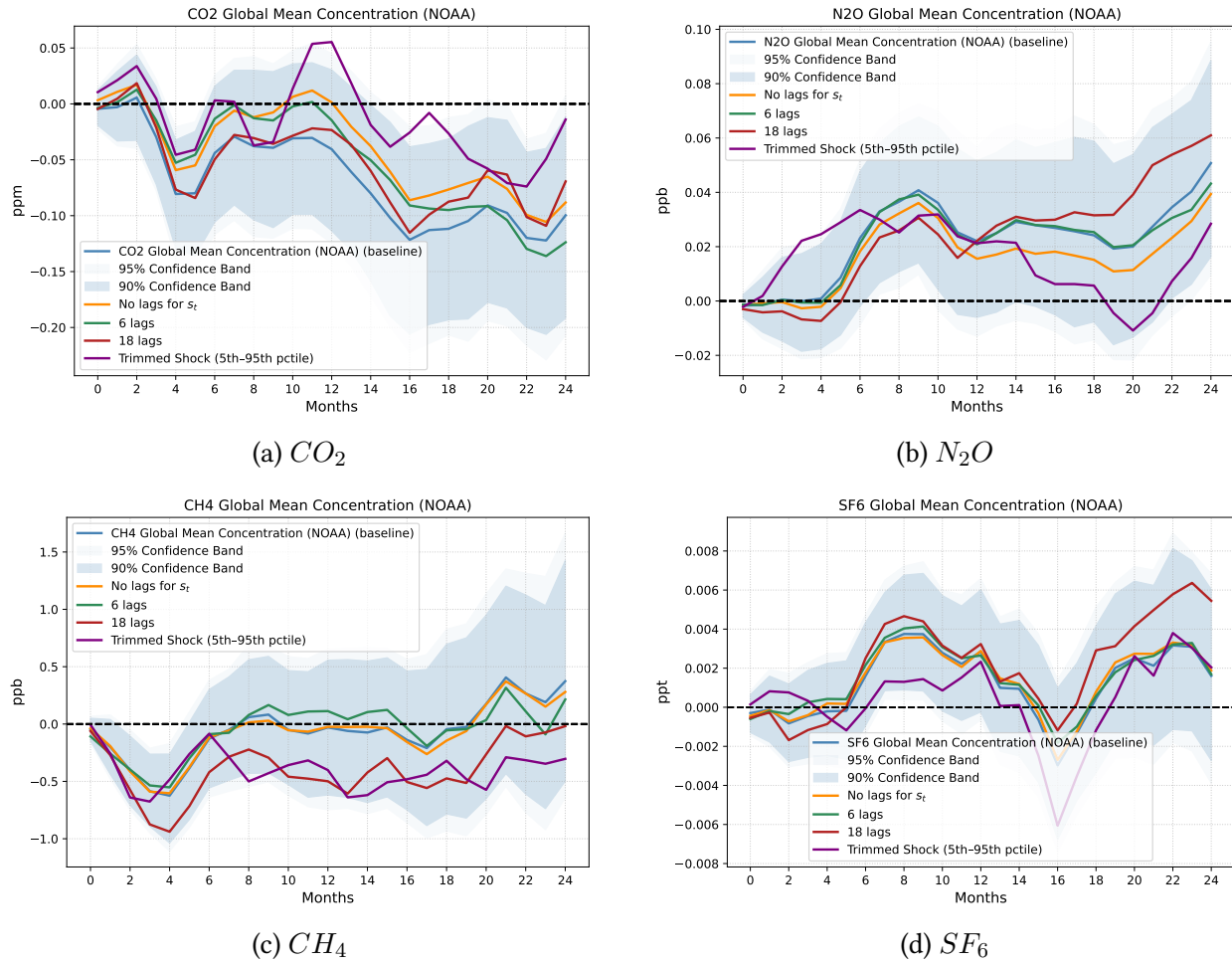
1. For each bootstrap replication $b = 1, \dots, B$, draw an independent multiplier $t_i^{(b)}$ for every observation i from the two-point distribution described below, and construct a bootstrap sample $y_i^{*(b)} = \hat{y}_i + t_i^{(b)} \hat{\varepsilon}_i$.
2. Re-estimate equation (1) on the bootstrap sample, holding the regressors fixed at their original values, to obtain $\hat{\beta}_h^{*(b)}$.
3. Repeat steps 1–2 for $B = 999$ replications, and construct the $(1 - \alpha)$ confidence interval for β_h from the $\alpha/2$ and $1 - \alpha/2$ percentiles of the empirical distribution of $\{\hat{\beta}_h^{*(b)}\}_{b=1}^B$.

Multiplier distribution. The multiplier t is drawn from Mammen's (Mammen 1993) two-point distribution, which satisfies $E[t] = 0$, $\text{Var}(t) = 1$, and $E[t^3] = 1$. The third-moment condition is what distinguishes this choice from a simple Rademacher (± 1 with equal probability) sign flip, and ensures that the bootstrap approximates the sampling distribution of $\hat{\beta}_h$ to a higher order of accuracy when that distribution is itself skewed. Concretely, t takes the value $-\frac{\sqrt{5}-1}{2}$ with probability $\frac{\sqrt{5}+1}{2\sqrt{5}}$, and the value $\frac{\sqrt{5}+1}{2}$ with probability $\frac{\sqrt{5}-1}{2\sqrt{5}}$.

D Robustness Results for Emissions

As described in subsection 2.4, I re-estimate equation (1) under four alternative specifications: no shock lags, 6 and 18 control and shock lags, and the shock trimmed to its 5th–95th percentiles. Figure D.1 plots the same robustness comparison for the remaining greenhouse gas series presented in figure 3: CO_2 , CH_4 , N_2O , and SF_6 , and confirms that the point estimates remain very similar.

Figure D.1: Robustness of Individual Greenhouse Gas Responses to Alternative Lag Specifications



Notes: Baseline impulse response (blue line, shaded 90%/95% confidence bands) to the oil supply news shock of Känzig (2021), overlaid with point estimates from four alternative specifications: no lags on the shock, 6 lags on both the target and the shock, 18 lags on both, and the shock trimmed to its 5th–95th percentiles. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

Asymmetric Response to Positive versus Negative Shocks To test for asymmetric dynamics, I augment equation (1) with an interaction between the shock and an indicator for a positive (price-increasing) realization:

$$\Delta^{h+1}y_{t+h} = \alpha_h + \alpha_{m(t),h} + \beta_h \cdot s_t + \gamma_h \cdot s_t \times 1_{\{s_t \geq 0\}} + \sum_{k=1}^{12} (\rho_h \Delta y_{t-k} + \Gamma'_h X_{t-k} + \tau_h s_{t-k}) \quad (6)$$

where $1_{\{s_t \geq 0\}}$ equals 1 when the shock is positive (an oil price increase) and 0 otherwise. The coefficient γ_h is the differential slope for positive shocks relative to negative ones at horizon h : for a unit shock ($s_t = \pm 1$), the response to a positive shock is $\beta_h + \gamma_h$, while the response to a negative shock is $-\beta_h$.

Figure D.2 plots γ_h , the coefficient on the positive-shock interaction term in equation (6) and the differential slope between positive (price-increasing) and negative (price-decreasing) oil supply news shocks, for both CO_2e and emissions efficiency.

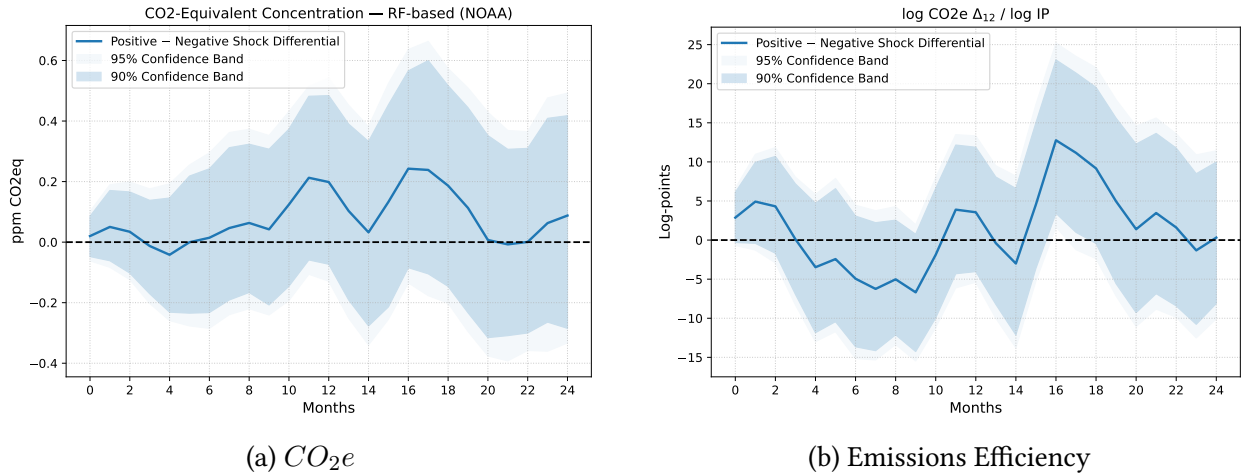


Figure D.2: Differential Response to Positive versus Negative Oil Price Shocks

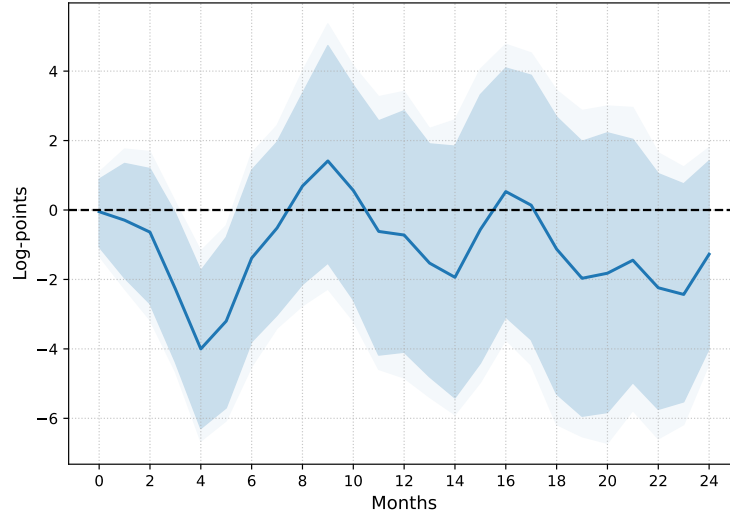
Notes: Point estimate and 90%/95% confidence bands for γ_h , estimated via equation (6). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

Emissions Efficiency Denominator Figure D.3 re-estimates the emissions-efficiency response of figure 2 using the 12-month growth rate of industrial production, rather than its level, as the denominator. The initial decline, concentrated in months 3–6, is nearly identical to the baseline.

E Panel Data Description

The empirical analysis draws on 6 main data sources. Table E.1 summarises the key characteristics of each dataset. For time-series variables, the table reports the sample span and number

Figure D.3: Emissions Efficiency Response — Industrial Production Growth-Rate Denominator



Notes: Impulse response to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in the log ratio of the 12-month difference in CO_2e concentration to the 12-month growth rate of industrial production. Shaded bands correspond to 90% and 95% confidence intervals based on heteroskedasticity-robust bootstrapped standard errors. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

of non-missing monthly observations. For panel variables it additionally reports the number of countries contributing with at least 1 non-missing observation over the sample period. All samples are restricted to January 2001 onwards to coincide with the availability of the NOAA N_2O concentration series — the binding constraint on the emissions sample, since the shock and control series all predate 2001. Oil supply news shocks run through December 2024. Panel outcome data extends to December 2025; the additional year contributes only as forward outcomes y_{t+h} for shocks realised in 2024, leaving the shock identification sample unchanged. Table E.2 reports start and end dates and summary statistics for the shock series and global control variables.

Electricity generation come from Ember, a monthly panel covering 85 countries from December 1998 onwards. Wind and solar installed capacity, also from Ember, is available for a much narrower sample of 25 countries starting in January 2016. Industrial production at the country level is sourced from the IMF, covering the OECD and selected emerging economies. Oil product consumption data (gasoline, diesel, jet fuel, LPG, and residual fuel) are from the Joint Organisations Data Initiative (JODI), as is natural gas consumption by the power sector. Coal production at the country level comes from the International Energy Agency (IEA). Coverage across these datasets is unbalanced: the number of countries reporting data varies across variables and over time, as documented in Figure E.1. Summary statistics for each variable used in the panel analysis are reported in Table E.3.

Table E.1: Data Sources and Summary Statistics

Dataset	Source	Countries	Start	End	Obs	Min	Mean	Median	Max	Std
CO ₂ e Concentration	NOAA	—	2001	2025	300	414.3	450.3	448.4	491.6	21.7
Real WTI Oil Price	FRED	—	2001	2025	299	0.2	0.8	0.7	1.7	0.3
Oil News Shock	KZ	—	2001	2024	288	-3.1	0.0	-0.0	1.9	0.6
Oil Supply Shock	BH	—	2001	2024	288	-1.9	0.1	0.1	5.3	0.6
Ind Prod	BH	—	2001	2025	300	60.0	85.0	85.1	110.7	14.2
Global Fin Cycle	MR	—	2001	2024	288	-2.9	0.2	0.1	2.6	1.0
Financial Uncertainty	JLN	—	2001	2025	300	0.7	1.0	1.0	1.4	0.1
Electricity Gen	Ember	85	2001	2025	10093	0.0	30.5	4.9	1015.2	94.7
Wind & Solar Capacity	Ember	25	2016	2025	2754	0.2	69.5	18.1	2082.1	185.1
(IMF) Ind Prod	IMF	65	2001	2025	16896	32.3	106.5	103.6	450.7	25.7
Oil Products	JODI	103	2002	2025	23474	0.0	7853.0	1209.0	306022.0	31392.2
Natural Gas	JODI	77	2009	2025	11866	0.0	2954.2	593.0	111046.0	8956.9
Coal Production	IEA	33	2016	2025	1188	0.0	65121.0	7768.0	1283191.0	183687.0

Notes: Sample restricted to years ≥ 2001 . COVID-19 (2020) is excluded from regressions but included in the statistics above. Obs. counts reflect non-missing observations for the key variable in each row. For panel variables, *Countries* reports the number of countries with at least one non-missing observation. KZ: Känzig (2021); BH: Baumeister and Hamilton (2019); MR: Miranda-Agrippino and Rey (2020); JLN: Jurado, Ludvigson, and Ng (2015).

Table E.2: Global Variables: Shocks and Controls

Variable	Source	Start	End	N	Mean	Std	Min	Median	Max
Oil Supply News Shock	Kaenzig (2021)	2001	2024	288	0.0	0.6	-3.1	-0.0	1.9
Structural Oil Supply Shock	Baumeister–Hamilton (2019)	2001	2024	288	0.1	0.6	-1.9	0.1	5.3
Global Financial Cycle	Miranda-Agrippino & Rey (2020)	2001	2024	288	0.2	1.0	-2.9	0.1	2.6
Financial Uncertainty (3M-ahead)	Jurado et al. (2015)	2001	2025	300	1.0	0.1	0.7	1.0	1.4
World Industrial Production (OECD+6)	Baumeister–Hamilton (2019)	2001	2025	300	85.0	14.2	60.0	85.1	110.7
Real WTI Oil Price	FRED	2001	2025	299	0.8	0.3	0.2	0.7	1.7

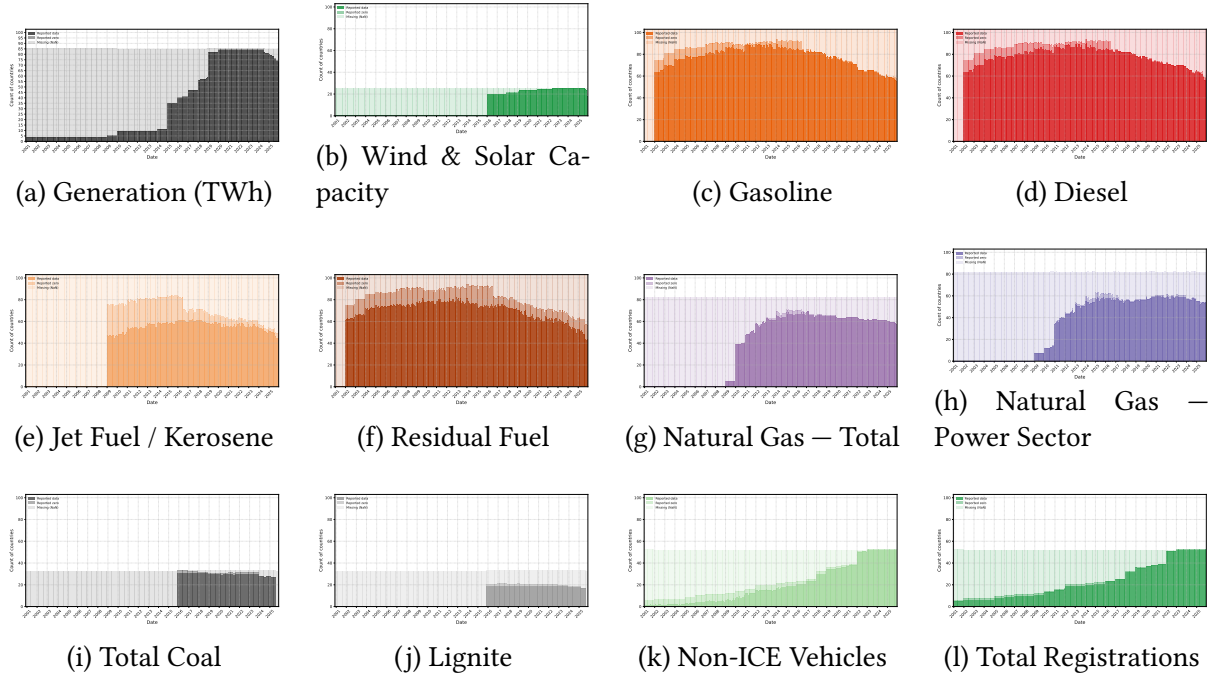
Notes: Sample restricted to years ≥ 2001 : *Start* is the later of that year and the series' first non-missing year, and *N*, Mean and all distributional statistics are computed over the same window. Oil supply shocks run through December 2024; outcome data extends to December 2025, with 2025 observations entering only as forward differences $y_{t+h} - y_{t-1}$ for shocks realised in 2024. KZ: Känzig (2021); BH: Baumeister and Hamilton (2019); MR: Miranda-Agrippino and Rey (2020); JLN: Jurado, Ludvigson, and Ng (2015).

Table E.3: Descriptive Statistics — Panel Variables

	Elec. Gen. (TWh)	W&S Cap. (GW)	IMF Ind. Prod.	ZEV Reg.	Alt+Hyb. Reg.	ICE Reg.	JODI Diesel	JODI Gasoline	JODI Jet Fuel	JODI LPG	JODI Res. Fuel	JODI Gas Power	JODI Gas Total	IEA Coal	IEA Cok. Coal	IEA Lignite
Mean	30.47	69.52	106.45	7,084.42	10,632.07	116,201.74	8,243.17	7,853.00	1,691.16	2,555.44	2,082.13	1,105.31	2,954.22	65,121.02	75,030.92	7,953.47
Std. Dev.	94.70	185.08	25.72	44,728.54	33,234.28	291,979.26	17,858.96	31,392.20	5,742.72	7,071.55	4,142.40	3,505.41	8,956.92	183,686.96	202,968.77	8,557.78
Min	0.02	0.20	32.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p25	1.53	6.40	93.02	16.00	96.25	6,992.25	634.10	346.45	15.76	81.20	53.00	50.33	242.00	620.00	240.50	1,016.00
Median	4.91	18.11	103.64	227.00	1,232.00	21,474.00	2,514.02	1,209.00	310.00	394.40	396.25	266.03	593.00	7,768.00	2,257.50	6,348.00
p75	17.53	50.58	115.48	2,064.75	6,309.50	91,977.75	7,451.77	4,599.26	1,086.41	2,041.60	1,949.84	799.25	2,911.00	31,914.50	66,272.00	11,520.00
Max	1,015.19	2,082.09	450.70	829,100.00	555,317.00	2,665,499.00	140,150.00	306,022.00	58,116.00	89,989.88	43,249.69	43,258.00	111,046.00	1,283,191.00	1,283,191.00	45,747.00
N	10,093.00	2,754.00	16,896.00	7,110.00	7,110.00	7,110.00	23,679.00	23,474.00	14,352.00	23,472.00	23,568.00	10,272.00	11,866.00	1,188.00	950.00	765.00
N Countries	85.00	25.00	65.00	52.00	52.00	52.00	103.00	103.00	91.00	103.00	103.00	78.00	77.00	33.00	30.00	22.00
N Periods	300.00	120.00	300.00	300.00	300.00	300.00	288.00	288.00	204.00	288.00	288.00	204.00	204.00	39.00	39.00	39.00
Max ctry./period	84.00	25.00	64.00	52.00	52.00	52.00	94.00	94.00	84.00	94.00	94.00	63.00	71.00	33.00	29.00	21.00
Min ctry./period	4.00	18.00	37.00	6.00	6.00	6.00	58.00	58.00	49.00	57.00	56.00	7.00	5.00	27.00	20.00	17.00
Start date	2001-01	2016-01	2001-01	2001-01	2001-01	2001-01	2002-01	2002-01	2009-01	2002-01	2002-01	2009-01	2009-01	2016-01	2016-01	2016-01
End date	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-12	2025-07	2025-07	2025-07

Notes: Elec. = Electricity; W&S = Wind & Solar; IMF Ind. Prod. = IMF Industrial Production Index; ZEV = Zero-Emission Vehicle; ICE = Internal Combustion Engine; Alt+Hyb. = Alternative Fuel + Hybrid; Cok. = Coking; Res. = Residual; JODI = Joint Organisations Data Initiative (quantities in KBBL = thousands of barrels, except gas power and total in M3 - cubic meters); IEA = International Energy Agency (quantities in KT = kilotons); Cry. = Countries. All the data are reported at a monthly frequency except for the IEA coal data which has a quarterly frequency.

Figure E.1: Country Coverage Across Panel Data Sources

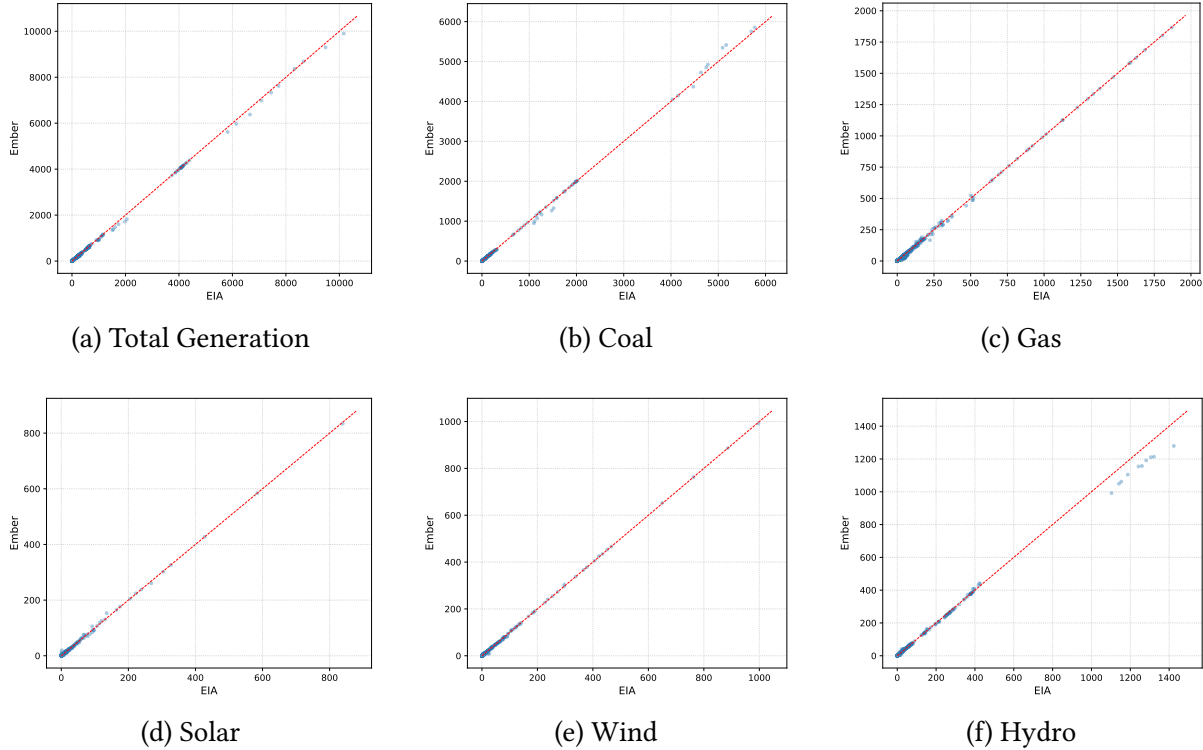


Notes: Each bar shows the number of countries with non-missing data in that month. Panels (a)–(b): electricity generation and wind & solar capacity (Ember). Panels (c)–(f): oil product consumption — gasoline, diesel, jet fuel/kerosene, and residual fuel (JODI). Panels (g)–(h): natural gas consumption, total and power sector (JODI). Panels (i)–(j): coal production, total and lignite (IEA). Panels (k)–(l): vehicle registrations, non-ICE and total. Each panel spans the full sample window, January 2001 – December 2025, regardless of each series' own reporting start date.

E.1 Ember–EIA Comparison

To validate the Ember dataset, I compare it against the annual international statistics published by the US Energy Information Administration (EIA). For each country and variable present in both datasets, I aggregate Ember’s monthly data to an annual total and compare it against the corresponding EIA annual figure. Figure E.2 plots this comparison for electricity generation by fuel type, and figure E.3 extends the comparison to installed wind and solar capacity.

Figure E.2: Ember vs. EIA Annual Electricity Generation



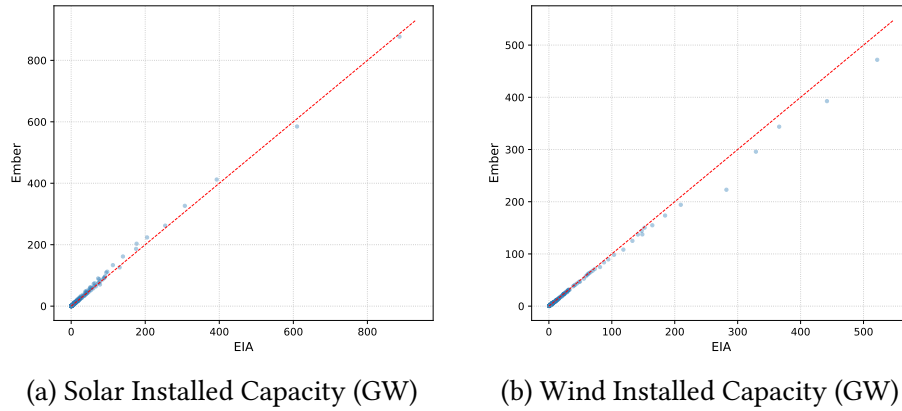
Notes: Each point represents 1 country-year observation. Ember monthly generation is aggregated to annual totals and compared against EIA annual reporting. Both axes in TWh.

F Panel Analysis Further Results

F.1 Total Coal Production

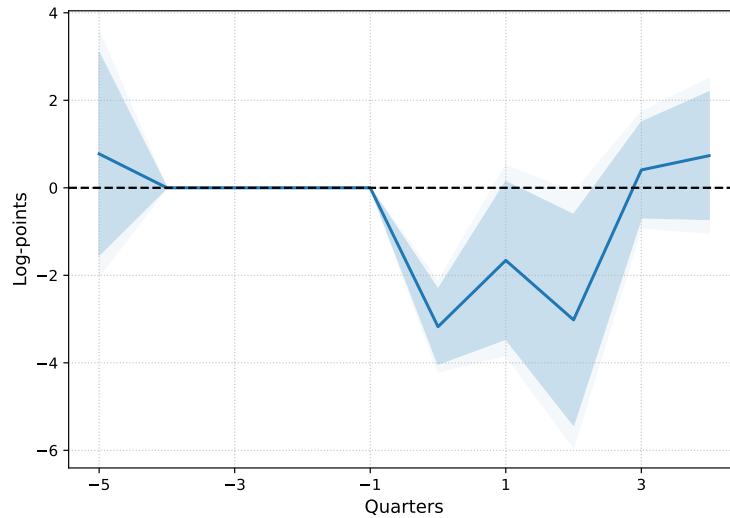
Figure 7 in the main text separates coal production by type. Figure F.1 reports the same specification for total coal production (steam, coking, and lignite combined), which decreases by 3.18 percentage points on impact and remains depressed for roughly half a year before recovering, following the dynamics of steam and coking coal.

Figure E.3: Ember vs. EIA Annual Installed Capacity



Notes: Each point represents 1 country-year observation. Ember annual capacity is compared against EIA annual reporting. Both axes in GW.

Figure F.1: Response of Total Coal Production to an Oil Supply News Shock

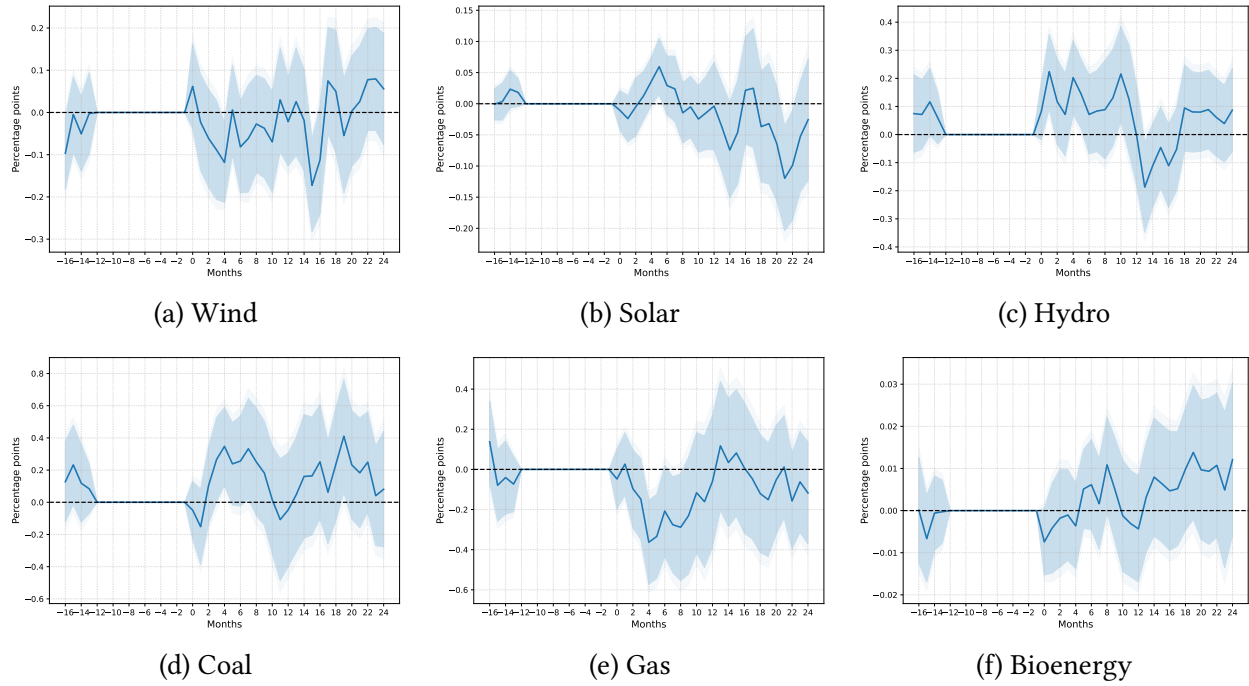


Notes: Impulse response to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log quarterly coal production; the horizon is measured in quarters. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-quarter. Regressions are weighted by the lag of the raw production level. COVID-19 (2020) is excluded from the sample. Shocks: January 2017 – December 2024; outcome data extends to July 2025.

F.2 Electricity Generation as a Share of Total

Figure F.2 reports wind, solar, hydro, coal, gas, and bioenergy generation as a percentage of total electricity generation, complementing the TWh-based responses reported in figures 7, 8 and 11.

Figure F.2: Electricity Generation by Source as a Percentage of Total Generation



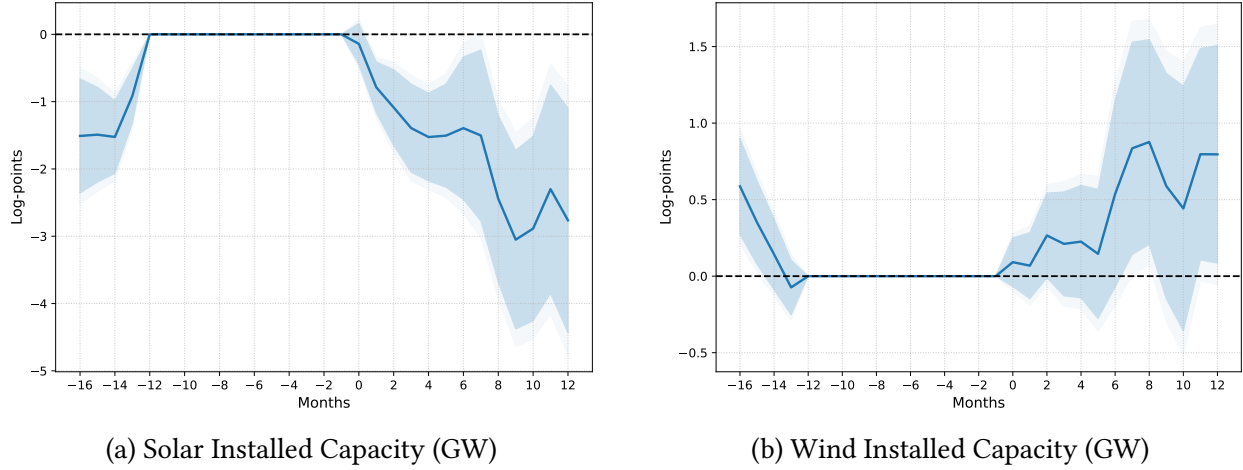
Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in the respective source's percentage share of total monthly electricity generation. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of total electricity generation (TWh). COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

F.3 Ember Capacity Data

The capacity panel covers 25 countries starting in January 2016 — a sample that is both geographically narrower and temporally shorter than the generation panel, which draws on 85 countries from January 2001. I present the impulse response functions for solar and wind nameplate capacity in figure F.3.

Any direct comparison between generation and capacity impulse responses risks conflating genuine causal effects with artefacts of sample composition: the 25 countries with capacity data may have systematically different generation dynamics, independent of any link between capacity and generation.

Figure F.3: Response of Renewable Installed Capacity to an Oil Supply News Shock

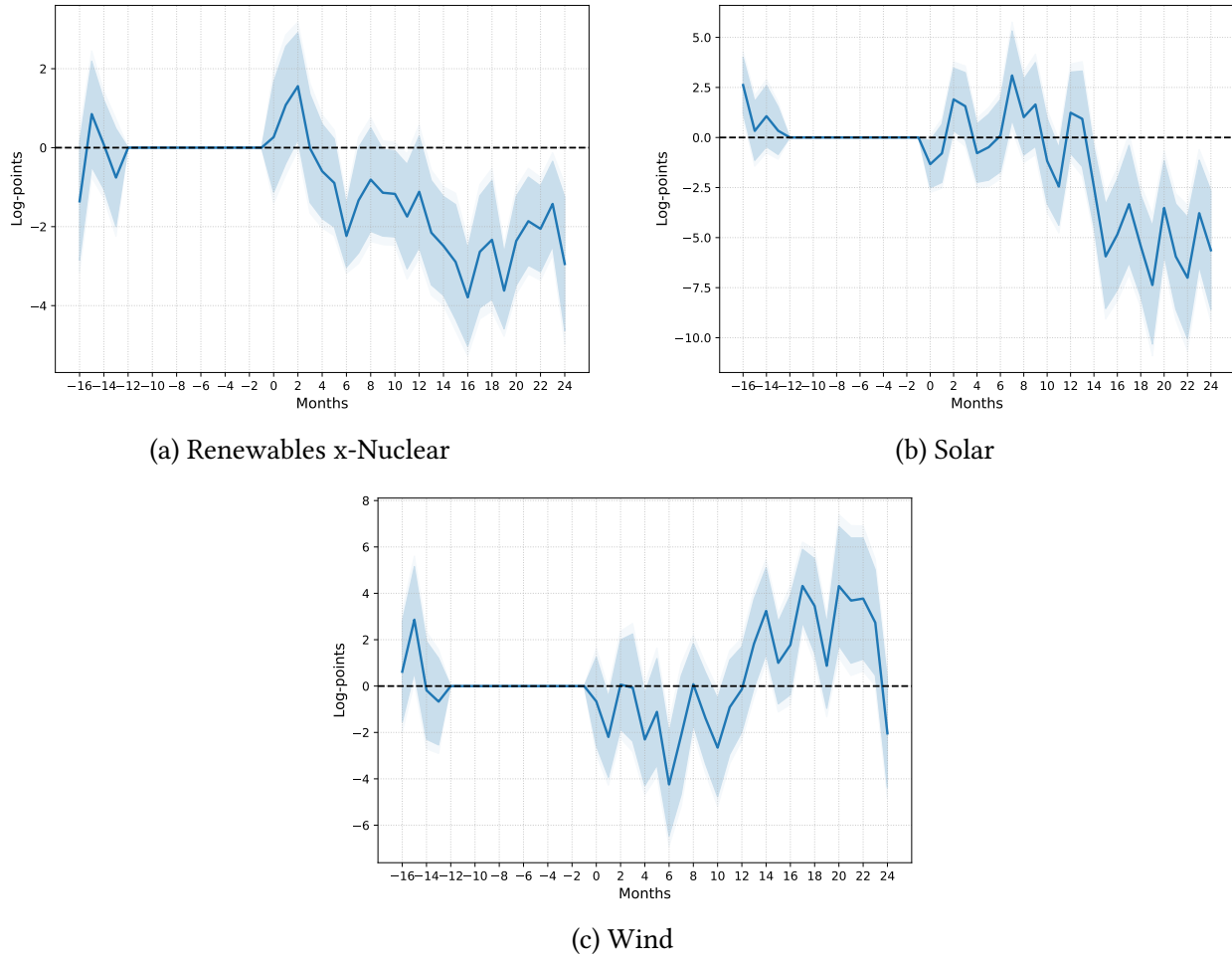


Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly installed renewable capacity (GW). Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective installed capacity (GW). Capacity data cover 25 countries. COVID-19 (2020) is excluded from the sample. Shocks: January 2017 – December 2024; outcome data extends to December 2025.

To address this concern, I re-estimate the generation local projections restricting the sample to those country-month observations for which the corresponding installed capacity variable is non-missing. Specifically, solar generation is estimated on the subsample defined by non-missing solar capacity, and wind generation on the subsample defined by non-missing wind capacity. This yields impulse responses on an estimation sample identical to those used for the capacity results in figure F.3, permitting a like-for-like comparison. The results are presented in figure F.4.

For solar, the dynamics closely replicate those from the full sample: the response follows the same trajectory stabilizing at log-points at the 2-year horizon, compared to log-points in the full sample. The solar generation decline is therefore not driven by differences in country coverage between the generation and capacity panels. For wind, the restricted sample produces a volatile response: generation initially declines but subsequently rises, with wide confidence bands throughout. This noisy pattern probably reflects the substantially smaller sample and the inability to capture the underlying signal. It also plausibly explains why wind capacity itself increases following the shock in figure F.3: on this same narrow set of countries, generation – not just capacity – also rises, suggesting the pattern reflects the composition of the capacity sample rather than a genuine investment response. The capacity data also has two further limitations worth noting: it does not account for depreciation, and it records only nameplate capacity, which may not reflect the actual potential power output of the installed capital.

Figure F.4: Response of Renewable Electricity Generation to an Oil Supply News Shock — Capacity Sample

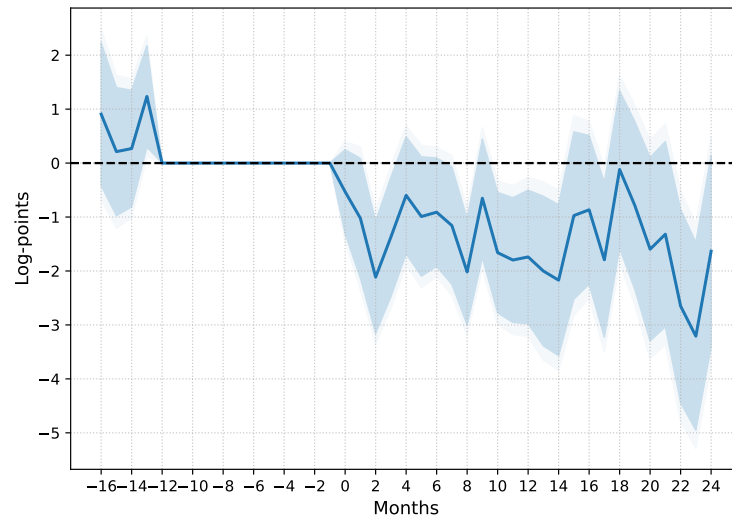


Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly electricity generation (TWh) from the respective source. Each regression is restricted to country-month observations for which the corresponding installed capacity variable is non-missing: solar generation and renewables-excluding-nuclear on the solar capacity sample, wind generation on the wind capacity sample. This matches the estimation sample to the capacity regressions reported in figure F.3. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective generation series (TWh). COVID-19 (2020) is excluded from the sample. Shocks: January 2017 – December 2024; outcome data extends to December 2025.

F.4 Vehicle Registrations

Total Registrations Figure 9 in the main text reports ICE and non-ICE vehicle registrations separately. Figure F.5 reports the same specification for total registrations (ICE and non-ICE combined). Total registrations fall by 2.02 percentage points at its trough, 8 months after the shock, and remain depressed after 2 years, 1.63 percentage points below trend.

Figure F.5: Response of Total Vehicle Registrations to an Oil Supply News Shock



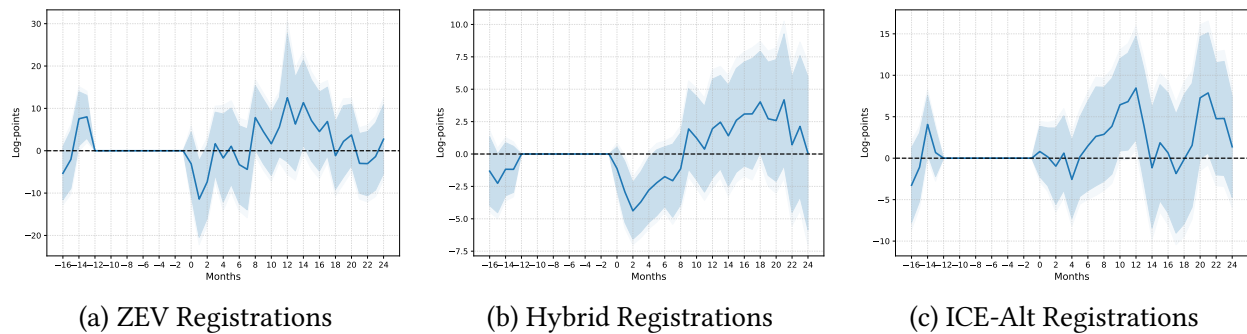
Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly vehicle registrations. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective vehicle registration count. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

Non-ICE Registrations Figure F.6 decomposes non-ICE registrations into zero-emission vehicles (ZEV), hybrids, and alternative-fuel ICE vehicles (ICE-Alt: LPG and flex-fuel). ZEV and hybrid registrations drive the aggregate pattern in figure 9: ZEV registrations fall 7.34 percentage points after 2 months and overshoot to 7.15 percentage points at month 15, while hybrids fall 4.39 percentage points before overshooting to 2.60 percentage points. ICE-Alt registrations barely fall on impact, by only 0.96 percentage points, but overshoot earlier, reaching 6.44 percentage points after 10 months.

F.5 Panel Robustness Checks

Every panel local projection reported in the main text (sections 3 and 4) is re-estimated under four alternative specifications, each targeting a distinct potential concern and each overlaid as a point estimate on the corresponding baseline figure:

Figure F.6: Response of Non-ICE Vehicle Registrations by Type to an Oil Supply News Shock



Notes: Impulse responses to the oil supply news shock of Känzig (2021), normalised to a 10% increase in oil prices on impact. The dependent variable is the $h + 1$ long-difference in log monthly vehicle registrations. Shaded bands correspond to 90% and 95% confidence intervals based on standard errors clustered by year-month. Regressions are weighted by the lag of the respective registration count. COVID-19 (2020) is excluded from the sample. Shocks: January 2002 – December 2024; outcome data extends to December 2025.

- *No lags for s_t* : drops the shock's own lags from the right-hand side of equation (2), to show that the results are not purely driven by noise.
- *Country + Month FE*: replaces the baseline's country-month fixed effects, which allow each country its own seasonal pattern, with separate, additive country and calendar-month fixed effects, imposing a common seasonal pattern across countries.
- *Alternative Weight (12-m Rol. Avg)*: replaces the baseline's single-period lagged weight with a 12-month rolling average of the same weight variable, to decrease the effect of seasonality in the weighting scheme itself, at the expense of potentially blurring genuine within-year changes in the true cross-country distribution of energy activity.
- *Trimmed Shock (5th–95th pctl)*: re-estimates using the oil news shock with values outside its own 5th–95th percentile band (computed over 2001–2025) set to missing, to ensure that no outlier shock realisations are driving the estimated dynamics.

Figure F.7 plots all four overlays for every panel series reported in the main text, with one exception: the *trimming shock* check is omitted for coal (steam/coking coal and lignite), whose sample is far smaller than the other panels (33 countries, mostly entering between 2016 and 2016), and would imply even fewer observations.

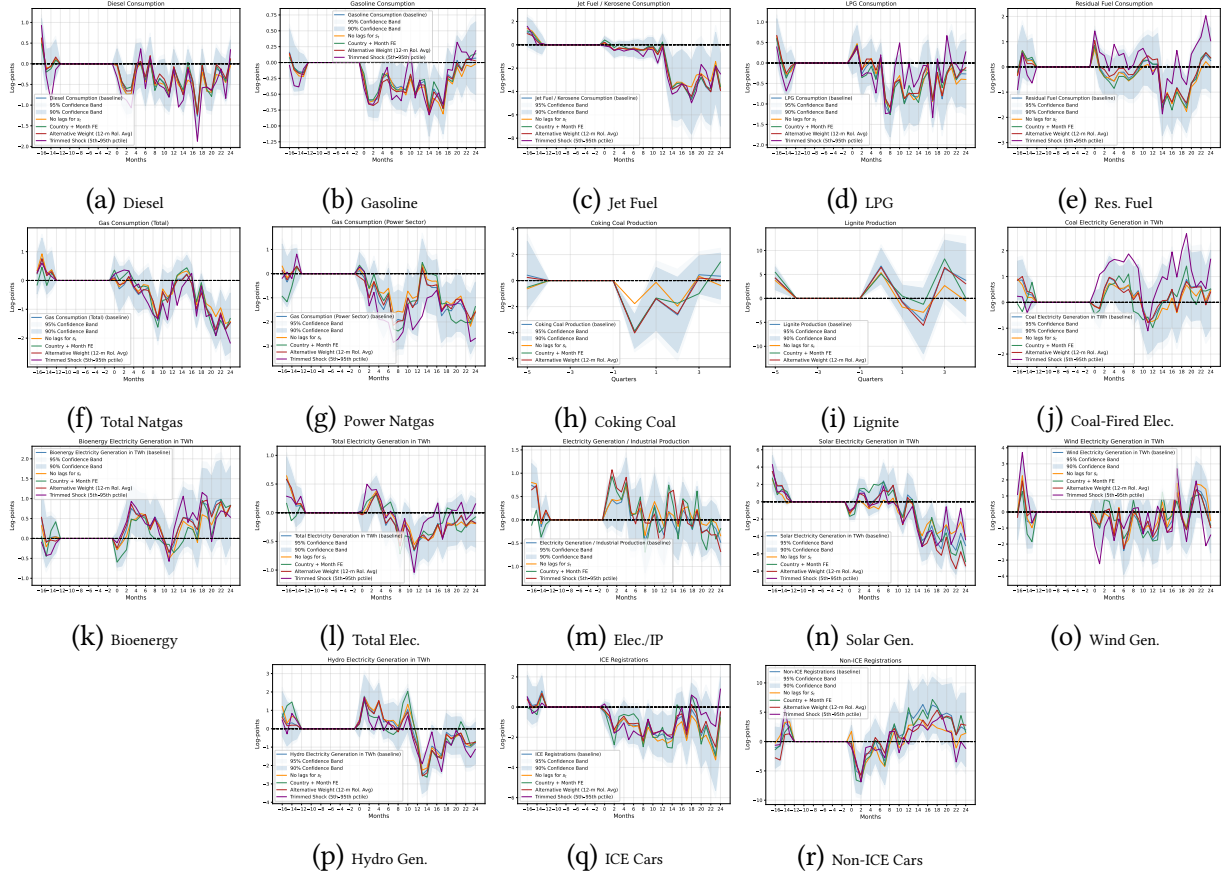


Figure F.7: Robustness of All Panel Local Projections to Alternative Specifications

Notes: Baseline impulse response (blue line, shaded 90%/95% confidence bands) to the oil supply news shock of Känzig (2021), overlaid with point estimates from the four alternative specifications described above, for every panel regression reported in the main text: refined oil consumption (figure 5), natural gas consumption (figure 6), coal production and coal-fired electricity generation (figure 7), total electricity generation and electricity-to-industrial-production (figure 10), bioenergy generation (figure 8), renewable generation by source (figure 11), and vehicle registrations (figure 9). COVID-19 (2020) is excluded from the sample.

References

- Baumeister, Christiane, and James D. Hamilton. 2019. Structural interpretation of vector autoregressions with incomplete identification: revisiting the role of oil supply and demand shocks. *American Economic Review* 109, no. 5 (May 1, 2019): 1873–1910. ISSN: 0002-8282. <https://doi.org/10.1257/aer.20151569>.
- Etminan, M., G. Myhre, E. J. Highwood, and K. P. Shine. 2016. Radiative forcing of carbon dioxide, methane, and nitrous oxide: a significant revision of the methane radiative forcing. *Geophysical Research Letters* 43 (24): 12, 614–12, 623. ISSN: 1944-8007. <https://doi.org/10.1002/2016GL071930>.
- Intergovernmental Panel on Climate Change (IPCC), ed. 2023. The earth’s energy budget, climate feedbacks and climate sensitivity. In *Climate change 2021 – the physical science basis: working group i contribution to the sixth assessment report of the intergovernmental panel on climate change*, 923–1054. Cambridge: Cambridge University Press. ISBN: 978-1-009-15788-9. <https://doi.org/10.1017/9781009157896.009>.
- Jurado, Kyle, Sydney C. Ludvigson, and Serena Ng. 2015. Measuring uncertainty. *American Economic Review* 105, no. 3 (March 1, 2015): 1177–1216. ISSN: 0002-8282. <https://doi.org/10.1257/aer.20131193>.
- Känzig, Diego R. 2021. The macroeconomic effects of oil supply news: evidence from OPEC announcements. *American Economic Review* 111 (4).
- Liu, Regina Y. 1988. Bootstrap procedures under some non-i.i.d. models. *The Annals of Statistics* 16, no. 4 (December): 1696–1708. ISSN: 0090-5364, 2168-8966. <https://doi.org/10.1214/aos/1176351062>.
- Mammen, Enno. 1993. Bootstrap and wild bootstrap for high dimensional linear models. *The Annals of Statistics* 21, no. 1 (March): 255–285. ISSN: 0090-5364, 2168-8966. <https://doi.org/10.1214/aos/1176349025>.
- Miranda-Agrippino, Silvia, and Hélène Rey. 2020. U.s. monetary policy and the global financial cycle. *The Review of Economic Studies* 87, no. 6 (November 1, 2020): 2754–2776. ISSN: 0034-6527. <https://doi.org/10.1093/restud/rdaa019>.
- Mori, Lorenzo, and Gert Peersman. 2024. *Estimating the macroeconomic effects of oil supply news*. SSRN Scholarly Paper 5054291. Rochester, NY: Social Science Research Network, November 1, 2024. <https://doi.org/10.2139/ssrn.5054291>.